LAXMI PRASOON BARIK

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Raipur, Chhattisgarh, India

Research Interest

- Adaptive Optics
- Exoplanets

Education

- **Ph.D.** *2026-current*
  Space, Planetary & Astronomical Sciences & Engineering, IIT Kanpur
- **Integrated M.Sc.** *2025*
  Center for Basic Sciences, PRSU,Raipur, C.G. | 84.90%
- **All India SSCE | 12th** *2020*
  Jawahar Navodaya Vidyalaya, Mahasamund, C.G. | 94.4%
- **All India SSE | 10th** *2018*
  Jawahar Navodaya Vidyalaya, Mahasamund, C.G. | 94.48%

Projects

- **Shack-Hartmann Based Atmospheric Turbulence Monitor** *Dissertation | January 2025*
  Advisor: Dr. Prashant Pathak, SPASE, IIT, Kanpur
- **Atmospheric Characterization Using Shack Hartmann Image Motion Monitor** *IAS-SRFP | September 2024*
  Advisor: Dr. Prashant Pathak, SPASE, IIT, Kanpur
- **Detection and Study of Characteristics of Quasars Using SDSS** *May 2024*
  Advisor: Dr. Laxmi Kant, Center for Basic Sciences, Raipur
- **Beyond Our Solar System: Discovering the Exoplanets** *December 2023*
  Advisor: Dr. Laxmi Kant, Center for Basic Sciences, Raipur
- **Atom Based Electric Field Sensing** *IAS-SRFP | August 2023*
  Advisor: Prof. Sudheesh Chethil, IIST, Thiruvananthapuram

Fellowships and Scholarships

- ***IIT Kanpur Institute Fellowship*** *2026-current*
- ***Center for Basic Sciences Student Scholarship*** *2020-25*
- ***IAS-INSA-NASI Summer Research Fellowship, 2024*** *July-September, 2024*
- ***IAS-INSA-NASI Summer Research Fellowship, 2023*** *July-August, 2023*

Skills

- **Programming:** Python, C, FORTRAN, MATLAB, Mathematica
- **Operating System:** Linux, Windows
- **Software and Typesetting:** GNU Plot, LaTeX, LibreOffice
- **Experimental Skills:** Basic Telescope Handling, FITS File Handling
- **Languages:** Hindi, English, Odia

Accomplishments and Experience

- **Program on All India Radio** *01 April, 2024*
  Yuv Vani, Akashvani Raipur
- **Appointed as Secretary** *2022-23*
  Student Council, PRSU, Raipur
- **Regional Science Congress** *December 2019*
  JNV Ratibad, Bhopal
- **Regional Mathematics Olympiad, Mahasamund** *2018*
  Indian Association of Physics Teachers
- **Top 0.1% in Social Sciences** *2018*
  AISSE

Seminars , Workshops and Courses Attended

- **Two Days Workshop On LaTeXfor Technical Writing** *29-30th April, 2025*
  School of Computational and Integrative Sciences, JNU, New Delhi
- **An Introductory Workshop on Exoplanet Detection and Characterization** *04-06th April, 2025*
  Indian Institute of Technology Kanpur
- **Poster competition on One day national conference on Molecules to Materials** *23rd February, 2025*
  SoS in Chemistry, PRSU, Raipur
  Poster Title- Shack-Hartmann Based Atmospheric Turbulence Monitor
- **Five day National Workshop on LaTeXfor Mathematicians and Physicists** *20-24th March, 2023*
  Center for Basic Sciences, PRSU, Raipur
- **Online lecture series on Nonlinear Dynamics and Applications** *13-16th February, 2023*
  Indian Institute of Technology, Indore
- **One-Day Seminar on Intellectual Property Rights** *30th November, 2022*
  Center for Basic Sciences, PRSU, Raipur
- **One-Day Workshop on IYBSSD** *July, 2022*
  Center for Basic Sciences, PRSU, Raipur
  Title- International Year of Basic Sciences and Sustainable Development
- **Diploma in Quantum Computing and Programming** *October, 2021*
  QWorlds's introductory tutorial Bronze - Qiskit
- **National Webinar on Coexisting with COVID-19** *13th May, 2021*
  Govt. Arts and Commerce Girls College, Raipur

- **Seminars Attended**
  - First-ever discovery of a temperature giant exoplanet through direct imaging - Dr. Prashant Pathak (Department of SPASE, IIT Kanpur)
  - Multifarious aspects of D-region modeling during solar flares using sub-ionospheric ULF propagation science - Dr. Tamal Basak (Indian Centre for Space Physics, Kolkata)
  - Gravitational waves as a probe for fundamental physics and astronomy - Dr. Ajit Kumar Mehta (University of California at Santa Barbara)
  - Gravitational Lensing Recent Developments in Strong Lensing - Prof. Jasjeet Singh Bagla (IISER-M and NCRA-TIFR, Pune)
  - Space Weather: How the Sun Influences Planetary Habitability and Human Technologies - Prof. Dibyendu Nandi (IISER Kolkata)
  - Supernovae Old and Young: From Remanants to Transients - Prasiddha Arunachalam (University of California)
  - The Lunar Regolith: Unveiling Its Composition and Significance - Dr. Megha Bhatt (PRL, Ahmedabad)
  - Synthetic Heliophysics Mission to Demonstrate Event-Driven Spacecraft Operations - Dr. Miles Bengston (Aurora Engineering)
  - Multi-frequency VLBI observations of solar and non-solar transients - Mugundhan V. (Onsala Space Observatory, Chalmers)
  - Building Imagining Systems in Space Platforms - Dr. Avyarthana Ghosh (Tata Research Development and Design Centre)
  - Spectro-polarimetry of Earth-like Exoplanets : science and techniques - Dr. Bhavesh Jaiswal (Space Astronomy Group, UR Rao Satellite Cente, ISRO)
  - Unravelling Cosmic Symphony of Merging Compact Binaries - Dr. Bhooshan Uday Gadre (GRASP, Utrecht University and National Institute for Subatomic Physics, Nikhef)

- **Popular Science Talks Attended**
  - Aditya L1: India's Spaceborne Sun Chaser - Prof. Durgesh Tripathi (IUCAA, Pune) -**5th April, 2025**
  - Kya Ham Taron Ki Dhool Se Bane Hain? - Prof. S.K. Pandey (Former VC, PRSU, Raipur) -**22nd Feb, 2025**
  - Popli Distinguished Lecture Series on "Common Man Science" on "From Einstein to Enigma" - Prof. Manish Jain (IIT Gandhinagar) -**11th Nov, 2024**
  - Light and Life - Prof. Harshawardhan Wanare (IIT Kanpur) -**22nd Oct, 2024**
  - Basic Quantum Physics and Bell's Inequality - Prof. Ajoy Ghatak (Formerly IIT Delhi) -**09th Sept, 2024**
  - National Space Day Event (Antarikx2024) - Organized by the Department of SPASE and Aerospace Engineering, IIT Kanpur -**24th Aug, 2024**

Extra Curricular Activities

- **Speech Competition** *February 2024*
  BoB, Regional Office, Raipur | 3rd Prize
- **International Essay Contest for Young People** *August 2023*
  The Goi Peace Foundation
- **Drama on Environment Day** *June 2023*
  Center for Basic Sciences, PRSU, Raipur
- **Volleyball at University festival: Abhyudaya** *April 2023*
  Pt Ravishankar Shukla University, Raipur | 1st Prize
- **Interdepartmental Essay Writing Competition** *March 2023*
  Department of Yoga, PRSU, Raipur
- **Role-play Competition** *October 2022*
  Center for Basic Sciences, PRSU, Raipur | 1st Prize
- **State-level extempore on Yuva Mahotsava** *December 2019*
  Jagadalpur, C.G.

  Participation in various essay writing, elocution, arts, and rangoli making competitions during high school and graduation.

Extra Curricular Interests

- **Playing Flute**
- **Reading Novels**
- **Playing Volleyball and Table-Tennis**

References

1. **Dr. Laxmi Kant**
   Assistant Professor, Center for Basic Sciences,
   Pt Ravishankar Shukla University, Raipur
   Email: lkcphysicscbs@gmail.com

2. **Dr. Prashant Pathak**
   Assistant Professor, SPASE
   Indian Institute of Technology, Kanpur
   Email: ppathak@iitk.ac.in